

July 6, 2026

Editor-in-Chief
Remote Sensing of Environment

Dear Editor,

We are pleased to submit our manuscript, “How Well Do PEFT Methods Adapt Geospatial Foundation Models to Heterogeneous Inputs? A Systematic Evaluation on Prithvi-100M with Production Validation on a Chinese Domestic High-Resolution Corpus,” for consideration as a research article in *Remote Sensing of Environment*.

Scope fit. The manuscript reports a systematic empirical study of how parameter-efficient fine-tuning (PEFT) methods adapt the publicly available Prithvi-100M geospatial foundation model to heterogeneous remote sensing inputs, validates the laboratory benchmark on a 35,920-patch operational deployment over Linhe County, China, using 203 multi-sensor scenes (0.42–4.29 m ground sampling distance, 22 satellite families, six calendar quarters in 2025) with Esri 10 m global LULC as the supervisory signal, and finally replicates the same five-method protocol on the LoveDA urban/rural cross-domain benchmark to test an adapter-capacity hypothesis under explicit domain shift. This benchmark–deployment–cross-domain triple is, we believe, well aligned with RSE’s emphasis on environmental applications grounded in rigorous remote sensing methodology rather than method papers in isolation.

Principal contributions.

1. A controlled benchmark of five PEFT methods (linear probing, BitFit, LoRA at rank 8, Housby adapters, an input-stage GeoAdapter) across single-label classification (EuroSAT, 27 modality configurations), multi-label classification (BigEarthNet-S2), and semantic segmentation (LandCover.ai). 109 experiments in total.
2. A diagnostic finding that standard module-wise LoRA can silently miss Prithvi-100M’s PyTorch fused-attention projections because Q/K/V are packed into a single tensor. A split-QKV ablation isolates implementation failure from genuine method limitation.
3. Production validation on the AlphaEarth-System platform: Housby adapters deliver +0.116 absolute mIoU (+65.5% relative) over linear probing on the Linhe LULC corpus under a geographic (scene-level) split protocol. The Housby-over-linear ordering is preserved across EuroSAT, BigEarthNet-S2, LandCover.ai, Linhe, and LoveDA within the tested Prithvi-100M setting.
4. A complementary synthetic weak-label control showing that the PEFT delta on the same imagery is mediated by label source: brightness-thresholded weak labels yield only a +0.017 PEFT delta, while Esri-derived supervisory labels yield +0.116. This control is not an independent manual validation set, but it helps separate method behavior from label-source effects.
5. A 30-run cross-domain replication on LoveDA (5 methods $\times$ 2 directions $\times$ 3 seeds) reporting evidence consistent with an adapter-capacity hypothesis near 10^6 trainable parameters in this tested setting. Housby reaches +0.117 mIoU over linear probing under R $\rightarrow$ U transfer; every PEFT method below the threshold (linear probing, BitFit, LoRA $r=8/16$, GeoAdapter) plateaus within 10^{-3} mIoU of the all-background baseline despite a 7–9 $\times$ trainable-parameter advantage over linear probing. A per-

class diagnostic at $5\times$ learning rate and 60 epochs confirms that the smaller adapters recover at most a binary background-versus-foreground distinction rather than pure collapse or simple undertraining. We therefore present the capacity threshold as a Prithvi-100M/LoveDA diagnostic hypothesis, not as a general deployment rule.

Reproducibility. All training code, configuration files, per-seed log files, and the AlphaEarth-System platform that orchestrates the Linhe pipeline are released under an open-source license. The Linhe patch index (~ 35 k patches at 128×128 , RGB) is shareable subject to source-imagery licensing; the supervisory labels are entirely derived from the publicly available Esri 10 m LULC product.

Confirmation. This manuscript has not been published, accepted, or simultaneously submitted elsewhere. An earlier preprint version covering the public-benchmark portion only (Sections 1–8) was prepared for the CVPR 2026 EarthVision workshop but was not submitted following the workshop deadline; the present manuscript adds Section 9 (production validation on Linhe and cross-domain replication on LoveDA) and substantially revises the discussion. All experimental decisions and authorship are unchanged.

Suggested reviewers. We suggest, but defer to your judgment, reviewers with expertise in geospatial foundation models and remote sensing PEFT (e.g., authors of Prithvi-100M / SatMAE / Scale-MAE / Spectral-GPT), Chinese-domestic high-resolution remote sensing benchmarks, and operational LULC mapping.

We appreciate your consideration of our manuscript.

Sincerely,

Zhouning